

Nikolai Slavov

CONTACT INFORMATION

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Lab: slavovlab.net SCPC: center.single-cell.net
ORCID ID: [0000-0003-2035-1820](https://orcid.org/0000-0003-2035-1820)

EDUCATION

Princeton University, Princeton, USA

Ph.D. in Molecular and Quantitative Cell Biology **06.2010**

M.A. in Molecular Biology **02.2006**

Dissertation: Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations

Mentor: David Botstein

Massachusetts Institute of Technology (MIT), Cambridge, USA

B.S. in Biology **06.2004**

ACADEMIC APPOINTMENTS

Allen Institute, The Paul G. Allen Frontiers Group

Allen Distinguished Investigator **2020–present**

Northeastern University, Boston, USA

Associate Professor, Department of Bioengineering **2021–present**

Faculty Fellow, Barnett Institute **2017–present**

Assistant Professor, Department of Bioengineering **2015–2021**

Harvard University, Cambridge, USA

Postdoc in the Departments of Systems Biology and Statistics **2014–2015**

Massachusetts Institute of Technology (MIT), Cambridge, USA

Postdoc in the van Oudenaarden laboratory, Departments of Biology and Physics **2011–2013**

AWARDS & FELLOWSHIPS

✓ Allen Distinguished Investigator (\$ 1, 500, 000) **2020**

✓ Excellence in Mentoring Award **2018**

✓ NIH Director's New Innovator Award (\$ 2, 352, 750) **2016**

✓ Broad Institute of Harvard and MIT SPARC Award **2014**

✓ Princeton University Dean's Award **2010**

✓ IRCSET Postgraduate Research Fellowship (72, 000 €) **2007**

✓ Finalist in the Young European Entrepreneur Competition **2006**

✓ Princeton Graduate Fellowship (\$ 45, 000) **2004**

✓ MIT Undergraduate Fellowship (\$ 125, 000) **2001**

✓ Eureka Fellowship for Academic Excellence (10, 000 €) **2000**

✓ Bronze Medal in the 31st International Chemistry Olympiad (*IChO*) **1999**

✓ National Diploma for Exceptional Achievements in Chemistry **1999**

2021

2021

- ✓ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**✉
Multiplexed single-cell proteomics using SCoPE2
Nature Protocols (in press), DOI: [10.1101/2021.03.12.435034](https://doi.org/10.1101/2021.03.12.435034) [OA]
- ✓ **Slavov N**✉
Measuring Protein Shapes in Living Cells
Journal of Proteome Research, DOI: [10.1021/acs.jproteome.1c00376](https://doi.org/10.1021/acs.jproteome.1c00376)
- ✓ **Slavov N**✉
Increasing proteomics throughput
Nature Biotechnology, DOI: [10.1038/s41587-021-00881-z](https://doi.org/10.1038/s41587-021-00881-z)
- ✓ **Slavov N**✉ *et al.*
Voices of biotech research: Single-cell proteomics
Nature Biotechnology, 39, 281286, DOI: [10.1038/s41587-021-00847-1](https://doi.org/10.1038/s41587-021-00847-1)
- ✓ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, Serra M, Kharchenko P, Koller A & **Slavov N**✉
Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2
Genome Biology, 22, 50 DOI: [10.1186/s13059-021-02267-5](https://doi.org/10.1186/s13059-021-02267-5) [OA]

2020

2020

- ✓ Specht H, **Slavov N**✉
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
Journal of Proteome Research, DOI: [10.1021/acs.jproteome.0c00675](https://doi.org/10.1021/acs.jproteome.0c00675)
- ✓ Petelski AA **Slavov N**✉
Analyzing ribosome remodeling in health and disease
Proteomics, 20, (17) :2000039 DOI: [10.1002/pmic.202000039](https://doi.org/10.1002/pmic.202000039)
- ✓ **Slavov N**✉
Single-cell protein analysis by mass-spectrometry
Current Opinion in Chemical Biology, 60, 1 - 9 DOI: [10.1016/j.cbpa.2020.04.018](https://doi.org/10.1016/j.cbpa.2020.04.018)
- ✓ **Slavov N**✉
Unpicking the proteome in single cells
Science, 367 (6477) :512 - 513, DOI: [10.1126/science.aaz6695](https://doi.org/10.1126/science.aaz6695)

2019

2019

- ✓ **Slavov N**✉ *et al.*
Voices in methods development: Single-cell proteomics
Nature Methods, 16, 945 - 951, DOI: [10.1038/s41592-019-0585-6](https://doi.org/10.1038/s41592-019-0585-6)
- ✓ Huffman RG, Chen AT, Specht H & **Slavov N**✉
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
Journal of Proteome Research, 18, 6, 2493 - 2500, DOI: [10.1021/acs.jproteome.9b00039](https://doi.org/10.1021/acs.jproteome.9b00039)
↔ [Technology feature](#) highlight by *Nature Methods*

- ✓ Chen AT, Franks A & **Slavov N** 
 DART-ID increases single-cell proteome coverage
PLoS Computational Biology, 5(7):e1007082 DOI: [10.1371/journal.pcbi.1007082](https://doi.org/10.1371/journal.pcbi.1007082) [OA]
 ⇨ [The single cell proteomics revolution](#), editorial highlight by *Bioanalysis Zone*
- ✓ Emmott EP, Jovanovic M & **Slavov N** 
 Approaches for studying ribosome specialization
Trends in Biochemical Sciences, 44(5):478 - 479, DOI: [10.1016/j.tibs.2019.01.008](https://doi.org/10.1016/j.tibs.2019.01.008)
- ✓ Emmott EP, Jovanovic M & **Slavov N** 
 Ribosome stoichiometry: from form to function
Trends in Biochemical Sciences, 44(2):95 - 109, DOI: [10.1016/j.tibs.2018.10.009](https://doi.org/10.1016/j.tibs.2018.10.009)
- ✓ Malioutov D., Chen T., Jaffe J., Airoidi E., Budnik B. & **Slavov N** 
 Quantifying homologous proteins and proteoforms
Molecular & Cellular Proteomics, 18(1):162 - 168, DOI: [10.1074/mcp.TIR118.000947](https://doi.org/10.1074/mcp.TIR118.000947)

2018

2018

- ✓ Specht H. & **Slavov N** 
 Transformative opportunities for single cell proteomics
Journal of Proteome Research, 17 (8), 2565 - 2571 , DOI: [10.1021/acs.jproteome.8b00257](https://doi.org/10.1021/acs.jproteome.8b00257)
 ⇨ [Innovations in Proteomics: The Drive to Single Cells](#), editorial highlight
 ⇨ [Through the Looking Glass of Single Cell Proteomics](#), highlight by *Technology Networks*
- ✓ Budnik B., Levy E., Harmange G. & **Slavov N** 
 SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, 19(1):161, DOI: [10.1186/s13059-018-1547-5](https://doi.org/10.1186/s13059-018-1547-5) [OA]
 ⇨ [SCoPE-MS – We can finally do single cell proteomics!!!](#), highlight by *Proteomics News*
 ⇨ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
 ⇨ [Single-cell proteomics](#) highlight by *the NIH Director's office*
 ⇨ [Technology feature](#) highlight by *Nature Methods*
 ⇨ [The single cell proteomics revolution](#), editorial highlight by *Bioanalysis Zone*
- ✓ Levy E. & **Slavov N** 
 Single cell protein analysis for systems biology
Essays in Biochemistry, 62(4):595 - 605, DOI: [10.1042/EBC20180014](https://doi.org/10.1042/EBC20180014)

2017

2017

- ✓ Franks A., Airoidi E.M., **Slavov N** 
 Post-transcriptional regulation across human tissues
PLoS Computational Biology, 13(5): e1005535, DOI: [10.1371/journal.pcbi.1005535](https://doi.org/10.1371/journal.pcbi.1005535) [OA]
 ⇨ [How Statistics Weakened mRNAs Predictive Power](#) highlight by *The Scientist*
- ✓ Saleh D, Najjar M, Zelic M, Shah S, Nogusa S, Polykratis A, Paczosa MK, Gough PJ, Bertin J, Whalen M, Fitzgerald KA, **Slavov N**, Pasparakis M, Balachandran S, Kelliher M, Meccas J, Degterev A.
 Kinase Activities of RIPK1 and RIPK3 Can Direct IFN- β Synthesis Induced by Lipopolysaccharide.
The Journal of Immunology, 198(11):4435-4447, DOI: [10.4049/jimmunol.1601717](https://doi.org/10.4049/jimmunol.1601717)

2016

2016

- ✓ Klionsky, *et al.*, **Slavov N**, *et al.*
Guidelines for the use and interpretation of assays for monitoring autophagy
Autophagy, vol. 12, issue 1, 1–222, 12(1):1-222, DOI: [10.1080/15548627.2015.1100356](https://doi.org/10.1080/15548627.2015.1100356)
- ✓ Di Luca A, Hamill RM, Mullen AM, **Slavov N**, Elia G.
Comparative Proteomic Profiling of Divergent Phenotypes for Water Holding Capacity across the Post Mortem Ageing Period in Porcine Muscle Exudate
PLoS One, 11(3):e0150605, DOI: [10.1371/journal.pone.0150605](https://doi.org/10.1371/journal.pone.0150605) [OA]

2015

2015

- ✓ **Slavov N** 
Making the most of peer review
eLife, 4:e12708, DOI: [10.7554/eLife.12708](https://doi.org/10.7554/eLife.12708) [OA]
- ✓ **Slavov N** , Semrau S., Airoidi E.M., Budnik B., van Oudenaarden A.
Differential stoichiometry among core ribosomal proteins
Cell Reports, vol. 13, issue 5, 865–873, DOI: [10.1016/j.celrep.2015.09.056](https://doi.org/10.1016/j.celrep.2015.09.056) [OA]
↔ [All Ribosomes Are Created Equal. Really?](#), highlight by *Trends in Biochemical Sciences*
↔ [New direction for tissue engineering and cancer therapeutics](#), highlight by *Science X*
- ✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N**, *et al.*
De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
Cell Metabolism, vol. 21, issue 5, 764–776, DOI: [10.1016/j.cmet.2015.04.003](https://doi.org/10.1016/j.cmet.2015.04.003)

Started Slavov Lab at Northeastern University, August 2015

2014

2014

- ✓ Malioutov D. , **Slavov N** 
Convex Total Least Squares
Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 [OA]
- ✓ **Slavov N** , Budnik B., Schwab D., Airoidi E., van Oudenaarden A. 
Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
Cell Reports vol. 7, issue 3, 705–714 [OA]

2013

2013

- ✓ **Slavov N** , Carey, J., Linse, S. 
Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
ACS Chemical Neuroscience, vol. 4, 601–612 [OA]

- ✓ **Slavov N**[✉], Botstein D.[✉]
Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Deregulation of Cell Size and Gene Expression
Molecular Biology of the Cell, vol. 24, issue 2, 157–168 [\[OA\]](#)

2012

2012

- ✓ **Slavov N**[✉], van Oudenaarden A.[✉]
How to Regulate a Gene: To Repress or to Activate?
Molecular Cell, vol. 46, issue 5, 551–552 [\[OA\]](#)
- ✓ **Slavov N**[✉], Airoidi E.M., van Oudenaarden A., Botstein D.[✉]
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 [\[OA\]](#)

2011

2011

- ✓ **Slavov N**[✉], Macinskas J., Caudy A., Botstein D.[✉]
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 [\[OA\]](#)
- ✓ **Slavov N**[✉], Botstein D.[✉]
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 [\[OA\]](#)

2010

2010

- ✓ **Slavov N**[✉]
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 [\[OA\]](#)
- ✓ **Slavov N**[✉], Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University [\[OA\]](#)
- ✓ Silverman SJ., **Slavov N**, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.[✉]
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 [\[OA\]](#)

2009 and before

2009 and before

- ✓ **Slavov N[✉]**, Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079-4084 (“In This Issue” article) [\[OA\]](#)
- ✓ Tagkopoulos I. [✉], **Slavov N[✉]**, Kung S. [✉] (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5th IEEE Symposium on Bioinformatics and Bioengineering (BIBE’05).

PATENTS

- **Slavov N[✉]**, Budnik B., Specht H., Levy E. Mass spectrometry technique for single cell proteomics, 16/251,039 (2019)

PUBLISHED

PREPRINTS

[GOOGLE SCHOLAR](#)

- ▷ Leduc A, Huffman RG, **Slavov N[✉]** (2021)
Droplet sample preparation for single-cell proteomics applied to the cell cycle
bioRxiv, DOI: [10.1101/2021.04.24.441211](#) [\[OA\]](#)
- ▷ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N[✉]** (2021)
Multiplexed single-cell proteomics using SCoPE2
bioRxiv, DOI: [10.1101/2021.03.12.435034](#) [\[OA\]](#)
- ▷ Specht H, **Slavov N[✉]** (2020)
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
bioRxiv, DOI: [10.1101/2020.08.24.264994](#) [\[OA\]](#)
- ▷ Petelski AA, **Slavov N[✉]** (2020)
Analyzing ribosome remodeling in health and disease
arXiv, [arXiv:2007.05839](#) [\[OA\]](#)
- ▷ **Slavov N[✉]** (2020)
Single-cell protein analysis by mass-spectrometry
arXiv, [arXiv:2004.02069](#) [\[OA\]](#)
- ▷ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, *et al.* & **Slavov N[✉]** (2019)
Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity *bioRxiv*, DOI: [10.1101/665307](#) [\[OA\]](#)
↪ [Over 1,000 SINGLE CELL PROTEOMES!](#), highlight by *Proteomics News*
↪ [Technology feature](#) highlight by *Nature Methods*
- ▷ Specht H, Emmott E, Koller A, **Slavov N[✉]** (2019)
High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
bioRxiv, DOI: [10.1101/665307](#) [\[OA\]](#)

- ▷ Huffman G, Specht H, Chen AT, **Slavov N** (2019)
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
bioRxiv, DOI: [10.1101/512152](https://doi.org/10.1101/512152) [OA]
- ▷ Specht H, Harmange G, Perlman DH, Emmott E, Niziolek Z, Budnik B, **Slavov N** (2018)
Automated sample preparation for high-throughput single-cell proteomics
bioRxiv, DOI: [10.1101/399774](https://doi.org/10.1101/399774) [OA]
- ▷ Chen A, Franks A, **Slavov N** (2018)
DART-ID increases single-cell proteome coverage
bioRxiv, DOI: [10.1101/399121](https://doi.org/10.1101/399121) [OA]
- ▷ Levy E. & **Slavov N** (2018)
Single cell protein analysis for systems biology
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26965v1](https://doi.org/10.7287/peerj.preprints.26965v1) [OA]
- ▷ Emmott EP, Jovanovic M & **Slavov N** (2018)
Ribosome stoichiometry: from form to function
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26991v1](https://doi.org/10.7287/peerj.preprints.26991v1) [OA]
- ▷ Specht H. & **Slavov N** (2018)
Routinely quantifying single cell proteomes: A new age in quantitative biology and medicine
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26821v1](https://doi.org/10.7287/peerj.preprints.26821v1) [OA]
- ▷ Malioutov D., Chen T., Jaffe J., Airoidi E., Carr S., Budnik B., **Slavov N** (2017)
Quantifying homologous proteins and proteoforms
bioRxiv, DOI: [10.1101/168765](https://doi.org/10.1101/168765) [OA]
- ▷ Berg P., Budnik B., **Slavov N**, Semrau S. (2017)
Dynamic post-transcriptional regulation during embryonic stem cell differentiation
bioRxiv, DOI: [10.1101/123497](https://doi.org/10.1101/123497) [OA]
- ▷ Budnik B., Levy E., **Slavov N** (2017)
Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
bioRxiv, DOI: [10.1101/102681](https://doi.org/10.1101/102681) [OA]
 - ↪ [SCoPE-MS – We can finally do single cell proteomics!!!](#), highlight by *Proteomics News*
 - ↪ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
 - ↪ [Interview](#), highlight by *Front Line Genomics*
- ▷ **Slavov N** (2015)
From differential transcription of ribosomal proteins to differential structure of ribosomes
PeerJ Preprint, DOI: [10.7287/peerj.preprints.1504v1](https://doi.org/10.7287/peerj.preprints.1504v1) [OA]
- ▷ Franks A., Airoidi E.M., **Slavov N** (2015)
Post-transcriptional regulation across human tissues
bioRxiv, DOI: [10.1101/020206](https://doi.org/10.1101/020206) [OA]

- ▷ Slavov N[✉], Botstein, D., Caudy A. (2014)
Extensive regulation of metabolism and growth during the cell division cycle
bioRxiv, DOI: [10.1101/005629](https://doi.org/10.1101/005629) [OA]
- ▷ Slavov N[✉], Semrau S., Airoidi E.M., Budnik B., van Oudenaarden A. (2014)
Differential stoichiometry among core ribosomal proteins
bioRxiv, DOI: [10.1101/005553](https://doi.org/10.1101/005553) [OA]

OPINIONS & POPULAR SCIENCE PUBLICATIONS

- ▷ Slavov N[✉] (2018) We Must Demand Evidence of Peer Review
The Scientist, [Link](#)
- ▷ Slavov N[✉] (2017) What was rate limiting was the idea, not the technology
Front Line Genomics, [Link](#)
- ▷ Slavov N[✉] (2015) Making the most of peer review
eLife, [Link](#)
- ▷ Slavov N[✉] (2014) Accomplishments Over Accolades!
The Scientist, [Link](#)
- ▷ Slavov N[✉] (2014) Discoveries lie hidden behind the façade of popular assumptions
Cell Reporter, [Link](#)
- ▷ Slavov N[✉] (2013) The Best Projects Are Least Obvious
The Scientist, [Link](#)
- ▷ Slavov N[✉] (2013) A Wonderful Christmas with Mysterious Oscillations
PubChase, [Link](#)
- ▷ Slavov N[✉] (2012) The Mission of MIT
The MIT Tech, [Link](#)

INVITED KEYNOTE TALKS

1. Increasing the sensitivity, reliability, reproducibility and throughput of single-cell proteomics
HUPO Reconnect 2021 World Congress, Human Proteome Organization **Nov.15.2021**
2. Nano proteomic sample preparation for single-cell protein analysis
International Symposium on Microscale Separations and Bioanalysis **Jul.12.2021**
3. Droplet sample preparation for single-cell proteomics applied to the cell cycle
2021 AOHUPO, Human Proteome Organization **Jun.30.2021**
4. Computational opportunities for single-cell proteomics
MaxQuant Summer School **Jun.25.2021**
5. Mapping the transcriptome and proteome of tissues in 3D at single-cell resolution
Plant Cell Atlas **Apr.08.2021**

6. Characterizing transcriptional and post-transcriptional regulation from single-cell variability
CELLINK Partnership Conference **Dec.11.2020**
7. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity ([link](#))
The 2020 HUPO World Congress, Human Proteome Organization **Oct.20.2020**
8. Single-cell proteomics by mass-spectrometry ([link](#))
Netherlands Proteomics Platform, Utrecht, Netherlands **Jan.17.2020**
9. Quantifying proteins in single cells at high-throughput
Single-cell Omics Symposium at Peking University **Oct.20.2019**
10. High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
European Single Cell Proteomics Conference, Institute of Molecular Pathology (IMP) **Aug.27.2019**
11. Quantifying proteins in single cells at high-throughput
The 28th International KOGO Annual Conference, South Korea **Sept.06.2019**
12. DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
MaxQuant Summer School, University of Wisconsin– Madison, [Video](#) **Jul.25.2019**

INVITED
DEPARTMENTAL
TALKS & VISITS

1. Single-cell proteomic and transcriptomic analysis of macrophage polarization
Department of Cell Biology, Duke University **Jan.14.2021**
2. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Peking University **Oct.11.2020**
3. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Department of Biochemistry, Boston University **Sept.14.2020**
4. Quantifying proteins in single cells at high-throughput
Center for Proteomics and Metabolomics, Leiden University **Jan.16.2020**
5. Quantifying proteins in single cells at high-throughput
Biotech Research and Innovation Center, University of Copenhagen **Jan.15.2020**
6. Coordination among metabolism and cell division
Boston Children Hospital & Harvard Medical School **Nov.14.2019**
7. Single-cell proteomics quantifies the emergence of macrophage heterogeneity
Department of Physiology, Johns Hopkins University School of Medicine **Nov.6.2019**
8. Understanding biological systems: In search of direct causal mechanisms ([link](#)) [Video](#)
Models, Inference & Algorithms Initiative (MIA), Broad Institute of MIT & Harvard **Apr.3.2019**
9. Understanding and controlling cell differentiation using single-cell mass-spectrometry ([link](#))
Department of Systems Biology (Theory Lunch), Harvard Medical School **May.18.2018**
10. Principles of ribosome-mediated translational regulation
The Center for Engineering in Medicine, Massachusetts General Hospital **Mar.17.2017**
11. Principles of ribosome-mediated translational regulation
Systems Biology Seminar, Boston University **Oct.13.2016**
12. Ribosome-mediated translational regulation: Direct methods for quantifying the ribosome code
Center for Systems Biology, Columbia University **Apr.20.2016**
13. Coordinating metabolism and ribosome-mediated translational regulation
Department of Biology, New York University **Apr.19.2016**

14. Quantifying protein isoforms ([link](#))
Models, Inference & Algorithms Initiative (MIA), Broad Institute **Sept.14.2015**

INVITED PRESENTATIONS FOR FUNDING RECOMMENDATIONS • Current state of single-cell proteomics, standards, potentials and quantitative benchmarks
Functional Proteomics, HubMap & Common Fund, NIH **Oct.8–9.2020**

• Single-cell proteomics
Helmsley charitable trust, Gut cell atlas meeting, NYC **Oct.28.2019**

• Transformative opportunities for single-cell proteomics
NHLBI Single-cell workshop, National Institutes of Health (NIH) **Oct.17.2019**

INVITED CONFERENCE TALKS 1. Droplet sample preparation for single-cell proteomics applied to the cell cycle
CZI Seed Network Webinar **Jul.20.2020**

2. Single-cell Proteomic and Transcriptomic Analysis of Macrophage Heterogeneity ([link](#))
HUPO World Congress **Oct.20.2020**

3. Single-cell protein analysis by mass-spectrometry ([link](#))
The future of single cell analysis, Earlham Institute **Feb.16.2020**

4. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity ([link](#))
Applied Bioinformatics in Life Sciences **Feb.14.2020**

5. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity ([link](#))
Cellular and Molecular Bioengineering Conference, BMES **Jan.3.2020**

6. Single-cell proteomics for studying regeneration and aging ([link](#))
MDI Biological Symposium for Regeneration and Aging **Oct.9.2019**

7. Mapping the Transcriptome and Proteome of Human Testis in 3D
CZI Human Cell Atlas Seed Networks Meeting **Jul.30.2019**

8. Transformative opportunities for single cell proteomics
The Greater Boston Mass Spectrometry Discussion Group **Dec.13.2018**

9. Understanding and controlling cell differentiation based on comprehensive mass-spec quantification of proteins and signaling in single cells
UPENN Single Cell Biology Symposium, [Video](#) **Nov.13.2018**

10. Chair for the Single Cell Proteomics session of HUPO
Automated sample preparation for high-throughput single-cell proteomics **Oct.3.2018**

11. Using single-cell proteomics to engineer directed cell differentiation
Bioengineering & Translational Medicine Conference **Sept.27.2018**

12. Analyzing single cell proteomics data
ETH, Biognosys Discovery Proteomics Seminar **Jul.7.2018**

13. Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
CSHL Meeting: Single Cell Analyses **Nov.10.2017**

14. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Annual Single Cell Analysis USA Congress, Boston **Oct.23.2017**
15. Quantifying protein heterogeneity during the differentiation of single stem cells
Global CESI-MS Symposium **Oct.06.2017**
16. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Festival of Genomics, Boston **Oct.04.2017**
17. From differential transcription of ribosomal proteins to differential structure of ribosomes
Genetics, Genomics and Beyond, Calico **Sept.16.2017**
18. Differential stoichiometry among core ribosomal proteins
CSHL Translational Control, Cold Spring Harbor Laboratory **Sept.08.2016**
19. Principles of ribosome-mediated translational regulation
Center for Interdisciplinary Research on Complex Systems **Apr.12.2016**
20. Coordination among cell growth, cell division and gene expression
Broad Institute of MIT and Harvard, YSB **Dec.9.2015**
21. Coordinating cell growth and metabolism with ribosome-mediated translational regulation
Northeastern University, Biology Department Colloquium **Oct.19.2015**
22. Direct quantification of highly homologous proteins by mass-spectrometry indicates physiological changes in the protein composition of the ribosome
Broad Institute of MIT and Harvard, CBBO **Jun.25.2014**
23. Variable stoichiometry among core ribosomal proteins
Broad Institute of MIT and Harvard **Jun.19.2014**
24. Quantification of translational regulation of homologous proteins with unprecedented accuracy
Harvard HMS Systems Biology Theory Lunch **Mar.21.2014**
25. Exponentially growing cells can support a constant growth rate by very different metabolic strategies
Fourth Annual Physical Sciences in Oncology Meeting, NCI **Apr.18.2013**
26. Experimentally identified trade-offs of aerobic glycolysis
Center of Cancer Systems Biology at Tufts **Dec.18.2012**
27. Trade-offs of aerobic glycolysis
Harvard HMS Systems Biology Theory Lunch **Sep.12.2012**
28. The dynamics of exponential cell growth at constant growth rate
MIT Physical Sciences-Oncology Center **Jun.11.2012**
29. Metabolic cycling without cell division cycling
Bauer Forum at Harvard FAS Systems Biology **Apr.15.2011**

CONFERENCE TALKS

1. Quantifying homologous proteins and proteoforms
US HUPO Conference **Mar.14.2016**
2. Variable stoichiometry among core ribosomal proteins
2014 ASCB/IFCB Annual Meeting **Dec.8.2014**
3. The dynamics of exponential growth
GSA Yeast Meeting, Princeton University **Aug.5.2012**

4. The dynamics of exponential growth
Broad Institute of Harvard and MIT **Apr.17.2012**
5. Inference of Sparse Networks with Unobserved Variables. Application to Gene Regulatory Networks
International Conference on Artificial Intelligence and Statistics **May.14.2010**

CURRENT EXTERNAL FUNDING (GRANTS)	✓ NIH Director's New Innovator Award (\$ 2, 352, 750)	2016-2021
	Sole PI , 100 % DP2GM123497, Ribosome-mediated translational regulation during stem cell differentiation	
	✓ Allen Distinguished Investigator Award (\$ 1, 500, 000)	2020-2023
	Sole PI , 92 %, Tracking proteome dynamics in single cells	
	✓ CZI Seed Networks (\$ 400, 000)	2019-2022
	Lead PI , 40 %, Mapping the Transcriptome and Proteome of Human Testis in 3D	
	✓ Merck Exploratory Science Center Fellowship (Approved, not awarded yet: \$ 341, 177)	2020-2021
	Sole PI , 100 %, Characterizing the protein networks underpinning macrophage polarization	
	✓ NIH/ NHGRI	2020-2025
	Co-investigator , R01HG011087, Single-cell direct RNA sequencing using electrical zero-mode waveguides and engineered reverse transcriptases	
	✓ NIH/NCI R21	2020-2022
	Co-investigator , R21CA246150, Discovering proteins that explain single-cell heterogeneity in fibrillar migration	

PENDING EXTERNAL FUNDING (GRANTS)	✓ Merck Exploratory Science Center Fellowship (\$ 242, 827)	2019-2020
	PI , 100 %, Developing next generation single-cell proteomics and using it to characterize the functional polarization of macrophages	
	✓ Sanofi iAward (\$ 196, 000)	2018
	PI , 100 %, Developing a technology platform for discovering biomarkers and drug targets	
	✓ NEU Tier I Award (\$ 50, 000)	2016
	PI , 50 %, Internal funding	
	✓ Broad Institute of Harvard and MIT SPARC Award (\$ 100, 000)	2014
	PI , 100 %	
	✓ IRCSET Postgraduate Research Fellowship (72, 000 €)	2007
	PI , 100 %	

TEACHING & ADVISING

Courses Taught

Teaching websites

Methods of Bioengineering: Mass-spectrometry analysis, Northeastern University **2020 – 2020**
Introduction to mass-spectrometry analysis, [Website with lectures](#)

Principles of Bioengineering , Northeastern University Helping introduce doctoral students to research, Website	2020 – 2020
Method and Logic in Systems Biology and Bioengineering , Northeastern University Discussion of seminal papers, Website	2019 – 2019
Mathematical Methods for Engineers , Northeastern University Introduction to major concepts and methods in linear algebra differential equations, Website	2016 – 2020
Method and Logic in Quantitative Biology , Princeton University Introduction to major ideas and concepts in quantitative biology based on primary literature	2009 – 2010
Residential Graduate Student at Forbes College , Princeton University Advised students; organized language tables, sport events and trips	2009 – 2010
Integrated Quantitative Introduction to Natural Sciences , Princeton University Interdisciplinary curriculum including physics, computer science, biochemistry, genetics, physiology and emphasizing quantitative problem solving and the connections among these disciplines	2006 – 2010
Terrascope , MIT Selected for two subsequent years to lead undergraduate student groups working on complex real-world problems	2002 – 2003
Biochemistry , MIT Taught precepts and helped undergraduate students prepare for examinations	2003 – 2004

Research associates at Northeastern University

[Alumni website](#)

Dr. Edward Emmott

Currently Tenure-track professor at Liverpool University

Dr. Toni Koller

Currently Director of Koch Proteomics Facility at MIT

Dr. David Perlman

Currently Senior Principal Scientist at Merck

Graduate Students at Northeastern University [Group members website](#)

Harrison Specht, Bioengineering PhD student

expected graduation: 2021

Aleksandra Petelski, Bioengineering PhD student

expected graduation: 2022

R. Gray Huffman, Bioengineering PhD student

expected graduation: 2022

Jason Derks, Bioengineering PhD student

expected graduation: 2022

Tianchi Chen, Bioengineering PhD student (currently in Sontag group)

expected graduation: 2022

Andrew Leduc, Bioengineering PhD student

expected graduation: 2023

Shiri Tsour, Bioengineering PhD student

expected graduation: 2023

Joseph Collura, Bioengineering MS student

expected graduation: 2022

Collin Gwilt, Bioengineering MS student

expected graduation: 2022

Undergraduate students at Northeastern University [Alumni website](#)

Ezra Levy	<i>Currently PhD student at Princeton University</i>
Guillaume Harmange	<i>Currently PhD student at University of Pennsylvania</i>
Erin Provost	<i>Currently Research Associate I at Atalanta Therapeutics</i>
Albert Chen	<i>Currently Research Associate the Broad Institute</i>
Kaely Gallagher	<i>Currently Engineer at Sana Biotechnology, Inc.</i>
Ryan Posey	<i>Currently Research Associate at Beth Israel Deaconess Medical Center</i>
Aanie Phillips	<i>Currently Medical student at Baylor College of Medicine</i>
Matthew Dowling	<i>Currently Northeastern MS student</i>
Levon Rodriguez	<i>Currently Northeastern undergraduate student</i>
Sebastian Hymson	<i>Currently Northeastern undergraduate student</i>

SERVICE: SCIENTIFIC LEADERSHIP	• Organizing the fourth Single-Cell Proteomics Conference (SCP2021)	Aug 16–18 2021
	• Member of a think tank on Functional Proteomics NIH (OD)	Sept 8–9 2020
	• Organized the third Single-Cell Proteomics Conference (SCP2020)	Aug 18–19 2020
	• Co-organized Learning Meaningful Representations of Life workshop at NeurIPS	Dec 13 2019
	• Member of a think-tank on single-cell analysis NIH (NHLBI)	Oct 17–18 2019
	• Organized the second Single-Cell Proteomics Conference (SCP2019)	June 10–12 2019
	• Chair for the fifth Annual Single Cell Analysis USA Congress	May 14 2019
	• Chair for the Single Cell Proteomics session , HUPO	Oct 1–3 2018
	• Organized the first Single-Cell Proteomics Conference (SCP2018)	June 9–10 2018
SERVICE: PROFESSIONAL ACTIVITIES	• Peer-review: Nature, Science, PNAS, Nature Molecular and Structural Biology, Molecular Biology of the Cell, PLoS, Nucleic Acids Research, Bioinformatics, Annals of Applied Statistics, Cell Reports, eLife, Nature Communications Publons Profile	
	• Editorial roles: Editor for eLife and PeerJ, bioRxiv Affiliate, Member of the editorial board of Journal of Proteome Research	
SERVICE: DEPARTMENT COMMITTEES	• Member of the Search Committee for Barnett Institute Director	2018–2020
	• Member of the Departmental Faculty Search Committee	2016–2020
	• Member of the Graduate Student Admissions Committee	2017–2020